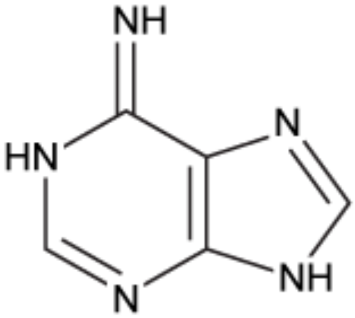
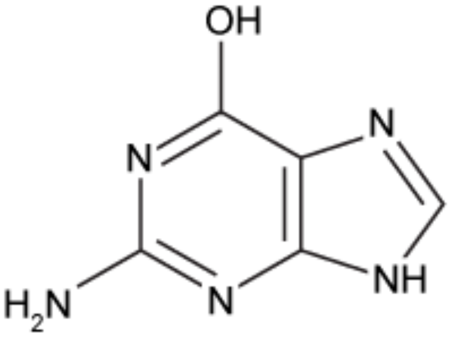
Which structure is a tautomer of guanine?

- ○ A.



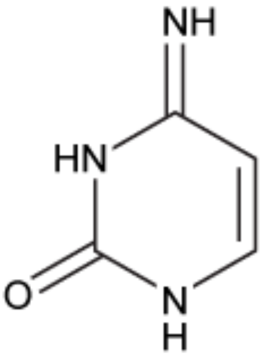
[17%]

- ◉ B. ✔



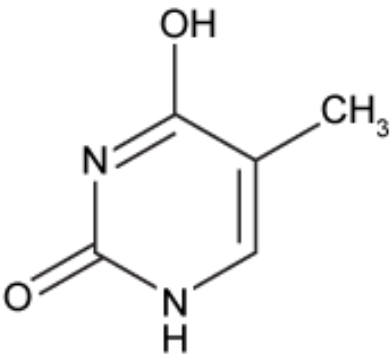
[71%]

- ○ C.



[5%]

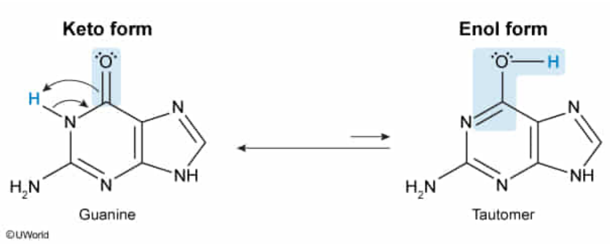
- ○ D.



[5%]

Correct
72% answered correctly

Explanation:



Tautomerization is a type of **isomerization** that involves the **transfer of hydrogen** from one position to another within a molecule and the **movement of a double bond** to an adjacent atom. These two forms, called tautomers, are in equilibrium. Tautomerization frequently involves a **keto** form (compound with C=O) as the major tautomer whereas the enol form is the minor tautomer. Tautomers are classified as constitutional isomers rather than resonance structures because a **bond is broken** during the hydrogen transfer to yield a different compound. On the other hand, resonance structures are forms of the same compound in which electrons move from one atom to another without breaking any sigma bonds.

The nitrogenous base guanine is a component of nucleotides and one of the four bases that are found in DNA. Guanine contains a δ-lactam and can be classified as a purine. It is a fused ring system with a ketone in the pyrimidine ring. Therefore, its tautomer must contain an enol.

(Choice A) This structure is a tautomer of the purine adenine, which contains an amine instead of a ketone. Therefore, its tautomer contains an **imine** (C=N) rather than an enol.

(Choice C) This structure is the tautomer of cytosine, which is a pyrimidine and contains only one nitrogenous aromatic ring. Like adenine, cytosine's tautomer is an imine.

(Choice D) This structure is a tautomer of thymine, which is a pyrimidine and contains a lactam. Like guanine, thymine contains a ketone, and its tautomer contains an enol.

Educational objective:
Tautomerization is a type of isomerization involving the transfer of a hydrogen atom from one position to another in a molecule and the movement of a double bond. The nucleotides can tautomerize, with guanine and thymine shifting between keto (major) and enol (minor) forms whereas adenine and cytosine shift between amines and imines.

Time spent:0 sec
Subject:
Organic Chemistry

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System:
Introduction to Organic Chemistry